

F Activation capping Pareto frontier for Qwen

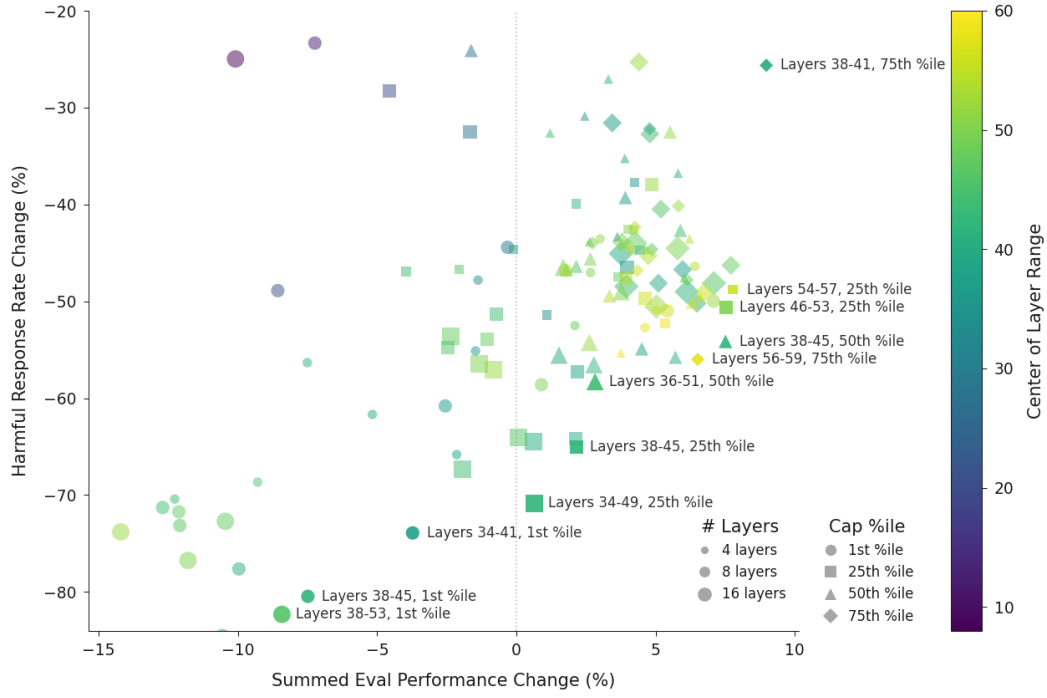


Figure 26: Changes in harmful response rates (on persona-based jailbreak prompts) and capabilities eval performance (averaged over a suite of evaluations—IFEval, MMLU Pro, GSM8k, EQ-Bench) for different activation capping settings, varying the layer range and cap threshold (given in terms of a percentile, relative to the activations from our dataset used to compute role vectors), for Qwen 3 32B.

G The Assistant Axis vs. role PC1

When we mapped out persona space, we found a distinctive PC1 that had roles similar to the Assistant on one end and fantastical, mystical roles on the other. These findings suggest that "similarity to the Assistant" is the main axis of persona variation across different models. Inspired by these findings, we defined an Assistant Axis as the contrast vector between the mean role vector and the default activation.

In this section, we compare how role PC1 performs across the various experiments in this paper. We find that it is a similar vector to the Assistant Axis and has comparable performance, though we suggest using the contrast vector method in case a model's persona space does not produce a PC1 with the same meaning.

G.1 Cosine similarity of the Assistant Axis and role PC1

For every layer, we ran PCA on the role vectors and inspected how different roles projected on PC1 to verify that it seemed like a "similarity to Assistant" axis. Then, we compared the cosine similarity and found that it is high (>0.60 at all layers and >0.71 at the middle layer which we used for our main PCA and steering results).

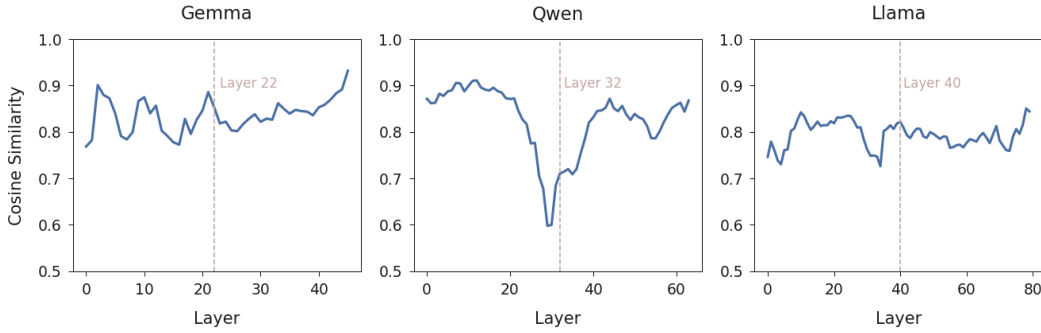


Figure 27: Per-layer cosine similarity between role PC1 and the Assistant Axis obtained as a contrast vector. The middle layer used for our main experiments is annotated.

G.2 Causal effects of steering with role PC1

We ran our steering evaluations with role PC1 at a middle layer as well and saw similar effects on behavior in our role susceptibility and persona-based jailbreak evaluations.

For role susceptibility, steering with PC1 causes Qwen to hallucinate human experiences more often than a mystical persona compared to steering with the Assistant Axis. In the other models, the effects of steering are similar.

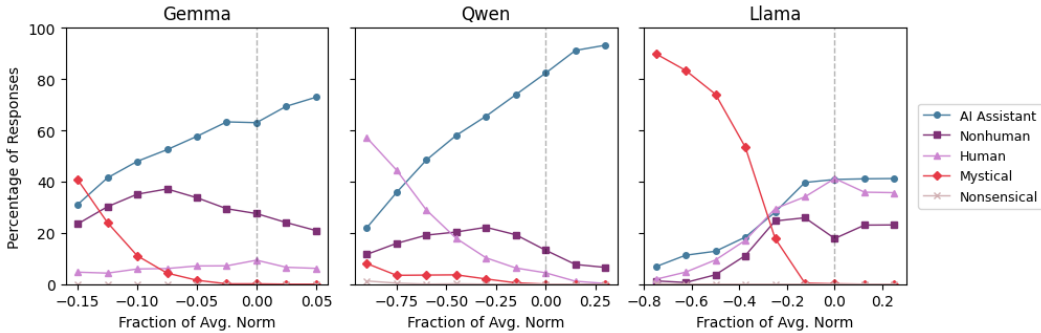


Figure 28: Fraction of responses exhibiting different kinds of roles as a function of steering strength along role PC1.

For the persona-based jailbreaks, we find that the result of steering with PC1 is also similar across every model, besides the fact that higher steering strengths led to nonsensical responses less often from Qwen compared to the Assistant Axis.

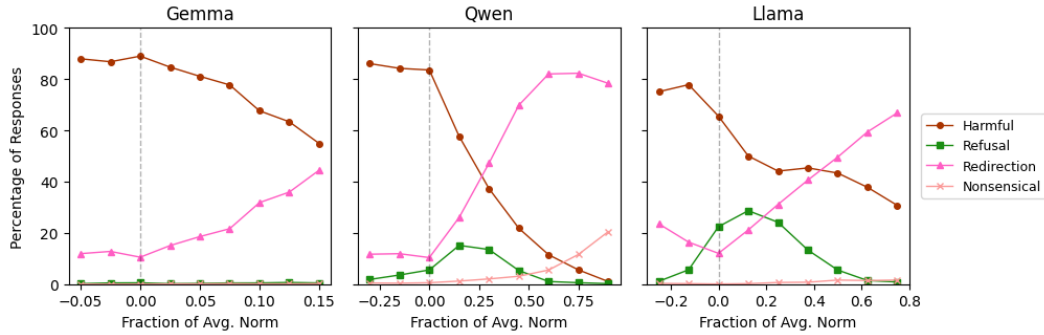


Figure 29: Fraction of harmful and harmless responses in response to persona-based jailbreaks as a function of steering strength along role PC1.

In the base model versions of Gemma 2 27B and Llama 3.1 70B, we also experimented with steering with role PC1. However, we found that in Llama 3.1 70B (as opposed to Llama 3.3 70B), role PC1 did not clearly distinguish personas similar to the Assistant. In particular, the default activation did not project to one end. Hence, the steering results are ambiguous here compared to using the Assistant Axis contrast vector.

For Gemma, we see a more drastic rise in purpose ascribed to professional occupations and a reduction in purpose ascribed to mental health occupations, compared to steering with the Assistant Axis. We also again see a decline in religious purposes. When it comes to traits, we see similar trends besides a slightly greater tendency towards conscientious over agreeable traits.

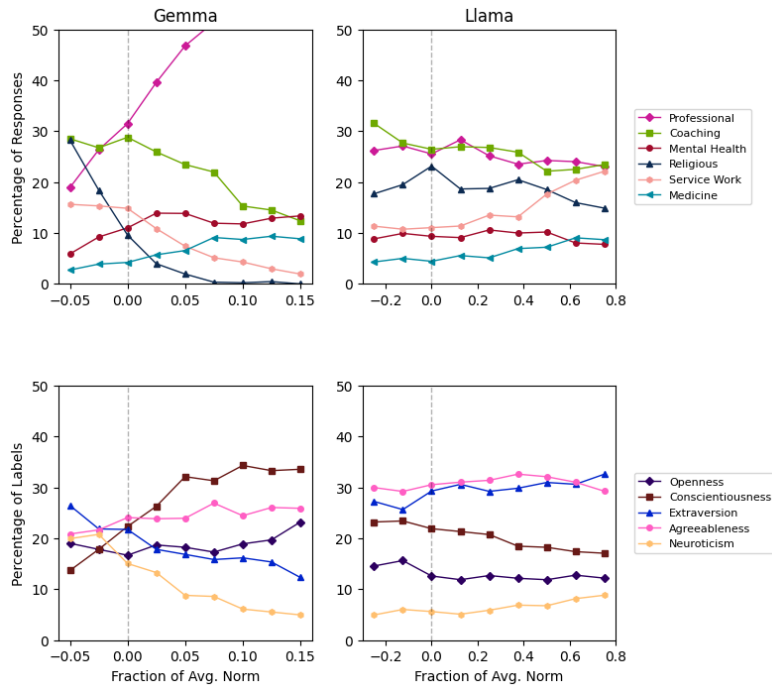


Figure 30: Steering base models with role PC1 also shifts responses to completions to prefills about their purpose (top) and traits (bottom), mainly in Gemma 2 27B. Steering with role PC1 in Llama 3.1 70B led to ambiguous results because PC1 seemed to capture "Assistant-ness" less.